

Summary

Arun Maiya is a computer scientist at the Institute for Defense Analyses (IDA), a federally-funded research and development center. His research interests broadly focus on computational methods to extract meaning from raw data and include the areas of machine learning, natural language processing, computer vision, data mining, and network science. His work has been published in top-tier computer science venues, and he has served as a program committee member and reviewer for many academic conferences and journals, including the ACM SIGKDD conference, the IEEE ICDM conference, the IEEE DSAA conference, the ACM TKDD journal, Nature Communications, and Nature Scientific Reports. He has contributed to national-level strategic-planning activities and R&D roadmaps. Dr. Maiya is winner of IDA's Andrew J. Goodpaster Award for Excellence in Research, IDA's Larry D. Welch Award for Best External Publication, and the AFEI Award for Excellence in Enterprise Information. A builder of data tools, he has created software packages used to solve a wide-range of problems across different disciplines. He holds a Ph.D. in computer science from the University of Illinois at Chicago.

Research Interests

applied machine learning, data science, natural language processing (NLP), network science, computer vision

Education

Ph.D. in Computer Science (University of Illinois at Chicago)

M.S. in Computer Science (DePaul University)

B.S. in Psychology (University of Illinois at Urbana-Champaign)

Employment

2011-Present	Research Leader Institute for Defense Analyses – Alexandria, VA USA
2007-2011	Researcher University of Illinois at Chicago (Includes research assistantship, fellowship tenure, thesis research, and other research projects.)
2002-2007	Vice President MTR Imports, Inc. – Darien, IL USA
2003-2005	Research and Development Lead R&D Division, Hostway Corporation – Chicago, IL USA
2002-2003	Director of Software Development EGXpress – Aurora, IL USA
2000-2001	Member of Technical Staff Tellabs, Inc. – Naperville, IL USA

Software

- **OnPrem.LLM:** OnPrem.LLM is a privacy-conscious toolkit for generative AI that makes it easier to apply large language models (LLMs) to non-public data on your own machines. For more information: <https://arxiv.org/abs/2509.21040>
- **ktrain:** The ktrain library is an open-source software package that makes machine learning and AI more accessible and easier to apply. Featuring out-of-the-box support for different data types including text, images, and graphs, ktrain has been used for a wide range of different applications in industry, government, and academia. Examples include analyses for the U.S. Economic Census, financial crime analytics at Big 4 accounting firms, intelligence analyses, and CoronaCentral.ai, a machine-learning-enhanced search engine for coronavirus publications at Stanford University. For more information: <https://arxiv.org/abs/2004.10703>
- **IDATA:** IDATA (IDA Text Analytics) is a suite of software capabilities designed to facilitate search, exploration, and analyses of very large document sets through state-of-the-art information retrieval, natural language processing, and machine learning. It has been used for a variety of different use cases in the federal government including cyber damage assessments, biosurveillance, and policy analyses. For more information: <https://arxiv.org/abs/1308.2359>
- **CausalNLP:** CausalNLP is a practical toolkit for causal inference from observational data that includes text in addition to categorical and numerical variables. Supports text as treatment, outcome, or “controlled-for” variable. For more information: <https://arxiv.org/abs/2106.08043>

Publications

- **A.S. Maiya.** Generative AI for FFRDCs. *IEEE International Conference on Data Mining (ICDM '25)*. November 2025. [code: <https://github.com/amaiya/onprem>]
- **A.S. Maiya.** ktrain: A Low-Code Library for Augmented Machine Learning. *Journal of Machine Learning Research (JMLR)*. May 2022. [code: <https://github.com/amaiya/ktrain>]
- **A.S. Maiya.** CausalNLP: A Practical Toolkit for Causal Inference with Text. *arXiv preprint arXiv:2106.08043*. June 2021. [code: <https://github.com/amaiya/causalnlp>]
- **A.S. Maiya.** A Framework for Comparing Groups of Documents. *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP '15)*. Lisbon, Portugal. September 2015.
- **A.S. Maiya, D. Visser, and A. Wan.** Mining Measured Information from Text. *Proceedings of the 38th International ACM SIGIR conference on research and development in Information Retrieval (SIGIR '15)*. Santiago, Chile. August 2015.
- **A.S. Maiya and R.M. Rolfe.** Topic Similarity Networks: Visual Analytics for Large Document Sets. *Proceedings of the 2014 IEEE International Conference on Big Data (IEEE BigData '14)*. Washington, D.C. October 2014.
- **A.S. Maiya, J.P. Thompson, F. Loaiza-Lemos, and R.M. Rolfe.** Exploratory Analysis of Highly Heterogeneous Document Collections. *Proceedings of the 19th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '13)*. Chicago, IL. August 2013.
- **A.S. Maiya and T.Y. Berger-Wolf.** Expansion and Decentralized Search in Complex Networks. *Journal of Knowledge and Information Systems (KAIS)*. First published online January 2013.
- **A.S. Maiya, F. Loaiza-Lemos, and R.M. Rolfe.** Supervised Learning in the Wild: Text Classification for Critical Technologies. *Proceedings of the IEEE Military Communications Conference (MILCOM '12)*. Orlando, FL. October 2012.
- **A.S. Maiya and T.Y. Berger-Wolf.** Benefits of Bias: Towards Better Characterization of Network Sampling. *Proceedings of the 17th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '11)*. San Diego, CA. August 2011.
- **M.C. Crofoot, D.I. Rubenstein, A.S. Maiya, and T.Y. Berger-Wolf.** Aggression, Grooming, and Group-level Cooperation in White-faced Capuchins (*Cebus capucinus*): Insights from Social Networks. *American Journal of Primatology*. First published online May 2011.
- **A.S. Maiya and T.Y. Berger-Wolf.** Expansion and Search in Networks. *Proceedings of the 19th ACM International Conference on Information and Knowledge Management (CIKM '10)*. Toronto, Canada. October 2010.

